

Dr. Mohsen Dirbaz · Founder · Halifax, Nova Scotia, Canada · dirbaz.sharif@gmail.com

July 16, 2026

By electronic transmission

Prof. Amnon Shashua
President and Chief Executive Officer
Mobileye Global Inc.
Jerusalem, Israel

Re: Exploratory inquiry — enforcement below the stack: a hardware complement to RSS, and governed memory above it

Dear Prof. Shashua:

I am the founder of Ghost Autonomy, a pre-silicon research program based in Halifax, Canada, developing what we call an **enforcement-and-accountability layer for bounded autonomy**: a risk-monotone actuation boundary terminating in non-programmable analog hardware, an evidence-admissibility discipline applied *before* sensor fusion, and a constitutionally governed memory in which every change to a system's operating truth is a logged, challengeable event. The program's distinguishing practice is evidentiary: every claim in our corpus carries a graded standing — measured, proven, proposed, or notional — and no claim is permitted to exceed it. This letter observes the same rule.

This letter differs from the others we are sending, because Mobileye is the one program whose differentiation, like ours, is a formal safety model. RSS made explicit assumptions a commercial product; IEEE 2846 made them an industry standard; True Redundancy made validation economics an architectural argument; and your Compound-AI position states publicly what we also hold — that unconstrained end-to-end monoliths cannot be validated. We therefore write not to differentiate against RSS but to build beneath it. RSS is enforced in software, within the programmable stack it polices, on best-estimate assumptions about other agents. Our work begins where that ends, with two questions: what enforces the envelope when the stack itself is the fault — and what governs the memory of every change to the system's operating truth? As Chauffeur scales eyes-off deployment and your first-party robotaxi service approaches its announced 2027 launch, we believe the enforcement locus and the epistemics of the assumption set become the critical questions.

At this point in your program's trajectory, we believe the most consequential engineering object you own is the set of assumptions on which it will continue — and that this set now deserves explicit registration, physical enforcement, and independent audit.

The offer. Ghost is prepared to pivot its development to maximize compatibility with the Mobileye product ladder: our φ contract and admissibility rules published as an IEEE 2846-compatible assumption registry; our conformance instrument shaped as assessor-facing tooling for the OEM programs deploying your systems; and our guard-ECU interface work specified against EyeQ-based system boundaries, as an independent enforcement channel rather than a competing stack.

What Ghost brings — specifically, and at its true standing.

1. **A measured boundary primitive.** An electrically isolated analog safety veto with ~ 32 ns intervention latency, witnessed on FPGA **[MEASURED]** — deliberately the only latency figure our corpus reports as measured rather than projected.
2. **Proven envelope mathematics.** The risk-monotone actuation algebra in its box case: projection is coordinatewise clipping; enforcement is firmly non-expansive (1-Lipschitz); the two-stage clamp is exact in the feasible case **[PROVEN]**.
3. **A fully specified refusal chain.** Five stages (S0–S4), from trusted numerics through fidelity gating, layered safety filtering, and risk-monotone envelopes to the analog veto, with six acceptance criteria (G1–G6) restated as a conjunctive, fail-closed conformance gate — published as criteria, not achievements **[PROPOSED]**.
4. **An admissibility-before-fusion evidence discipline.** Sensors as witnesses under integrity, alignment, scope, and lineage gates; correlated witnesses counted once; inadmissible evidence excluded rather than averaged in — sensor-agnostic by construction **[PROPOSED]**.
5. **A constitutional memory doctrine.** Append-only canon, anti-silent-drift amendment semantics, verification burden that increases with residence time, and recall as a priced contract (Governed Knowledge Systems trilogy, July 2026) **[NOTIONAL]**.
6. **A self-auditing corpus.** An 89-page consolidated foundational corpus (May 2026); a 94-page technical documentation edition, v3.0 (June 2026); and a 367-page compendium whose provenance ledgers grade every claim — including a published coverage map (68 subcategories: 21 full, 28 high, 9 partial, 10 gap) and a formal proof-debt ledger (P-1–P-7) naming what remains unproven.
7. **Empirical provenance, honestly scoped.** Physics-informed reduced-order kernels on elliptic-integral/AGM primitives with machine-precision convergence **[PROVEN]**, validated at $R^2 \approx 0.78\text{--}0.87$ on 200+ experimental datasets within their demonstrated envelope **[MEASURED]** — in a non-automotive domain, with automotive transfer registered in our own ledger as an open validation obligation.

What this could be worth to your program — specifically.

1. **True Redundancy, extended to enforcement.** Your argument multiplies perception reliability through channel independence; an electrically isolated, non-programmable actuation boundary applies the same independence argument to the enforcement path itself.
2. **One-sided epistemics as a certifiable strengthening.** Where RSS assumptions are calibrated estimates, a trusted scalar contractually permitted to err only toward higher risk gives an assessor a monotone bound rather than a distribution — complementary to RSS, and expressible in 2846 terms.
3. **Assessor-facing tooling for your OEM customers.** The claim ledger and conjunctive conformance gate as instruments an accredited assessor can run, reducing the per-program validation burden your supply-chain position ultimately prices.
4. **Governed memory for OTA and fleet learning.** Anti-silent-drift and priced recall as the data-governance layer for eyes-off consumer fleets under ISO/PAS 8800 — the layer of the market matrix presently empty at every operator.

What Ghost is not. We have no fabricated silicon, no vehicle deployment, and no fleet data; our measured hardware evidence is presently a single number. We state this plainly because we

regard the statement itself as an asset — a partner adopting our layer inherits a provenance ledger, not a marketing narrative — and because it is precisely why a compatibility-first alignment now, while our interfaces are still being fixed, costs little and shapes much.

Our inquiry is deliberately modest: *is such an alignment of interest?* As a first, zero-obligation step, we would run our conformance instrument against a trace or scenario format of your choosing and return the results with their standings — an artifact your team keeps regardless of what follows. Our two strategy volumes and the published corpus are available on request; deeper documentation under mutual NDA.

Respectfully,

Dr. Mohsen Dirbaz
Founder, Ghost Autonomy
Halifax, Nova Scotia, Canada
dirbaz.sharif@gmail.com

Evidentiary standings used in this letter — **[MEASURED]** witnessed on hardware; **[PROVEN]** a mathematical result; **[PROPOSED]** specified and defended, not yet validated; **[NOTIONAL]** a research direction, quotable as intent only. It is our house rule that no statement may exceed its standing; we extend the same rule to this letter.